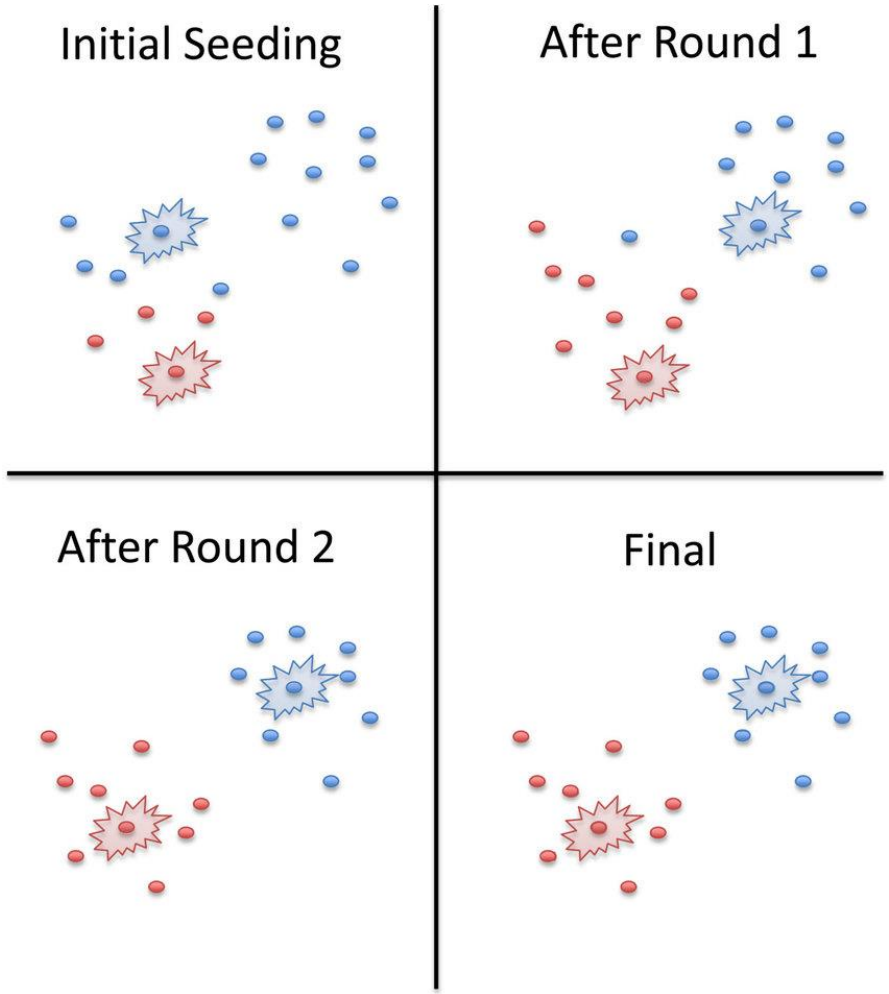
UTM

# Lab Tutorial using WEKA: Clustering using K-Means

Iris Dataset

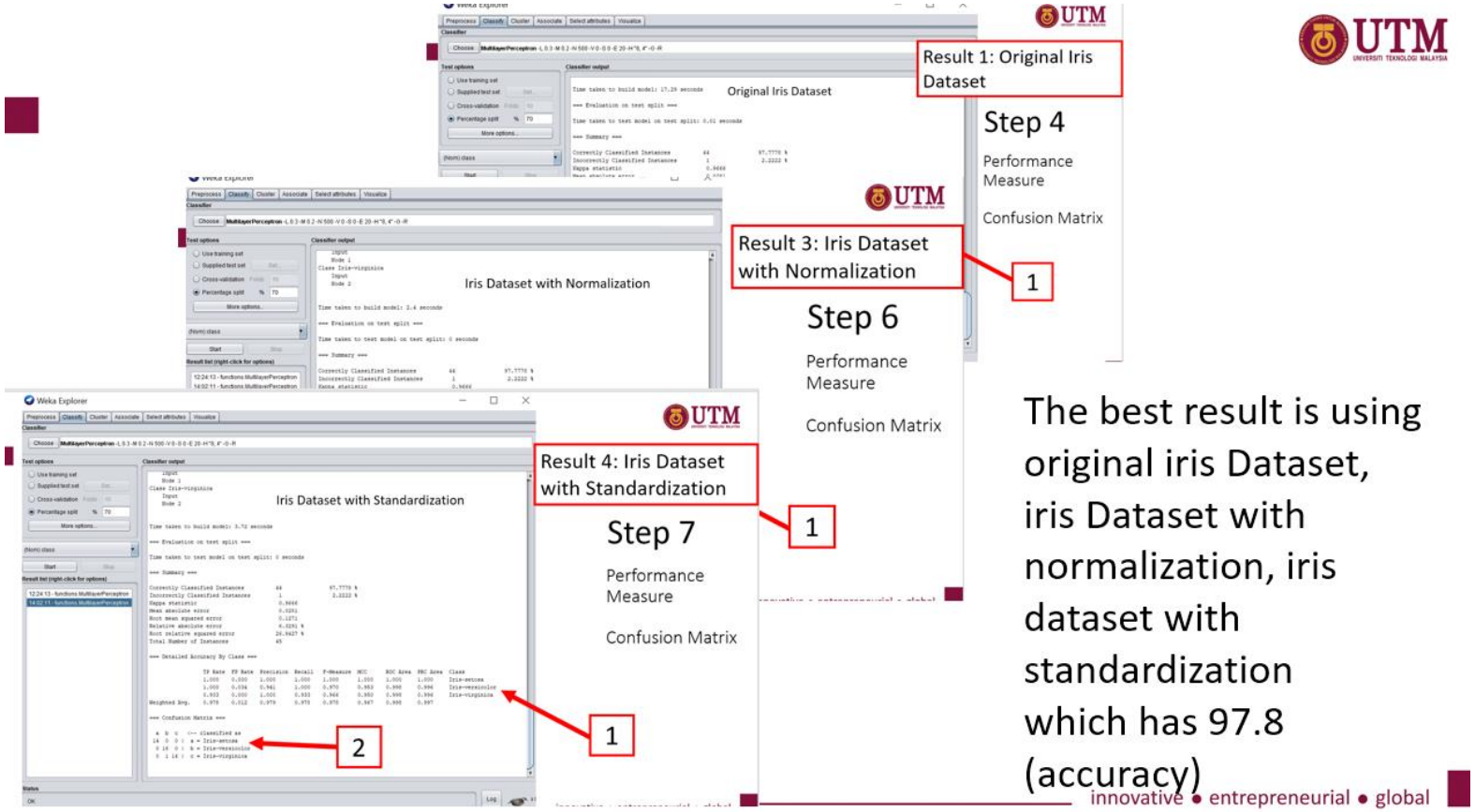
# Introduction to Clustering

- Clustering belongs to a group of techiques of unsupervised learning.
- It enables grouping instances into groups, where we know which are the possible groups in advance.
- These groups are called clusters.
- As the result of clustering each instance is being added a new attribute
- – the cluster to which it belongs. The clustering is said to be successful if the final clusters make sense, if they could be given meaningful names.

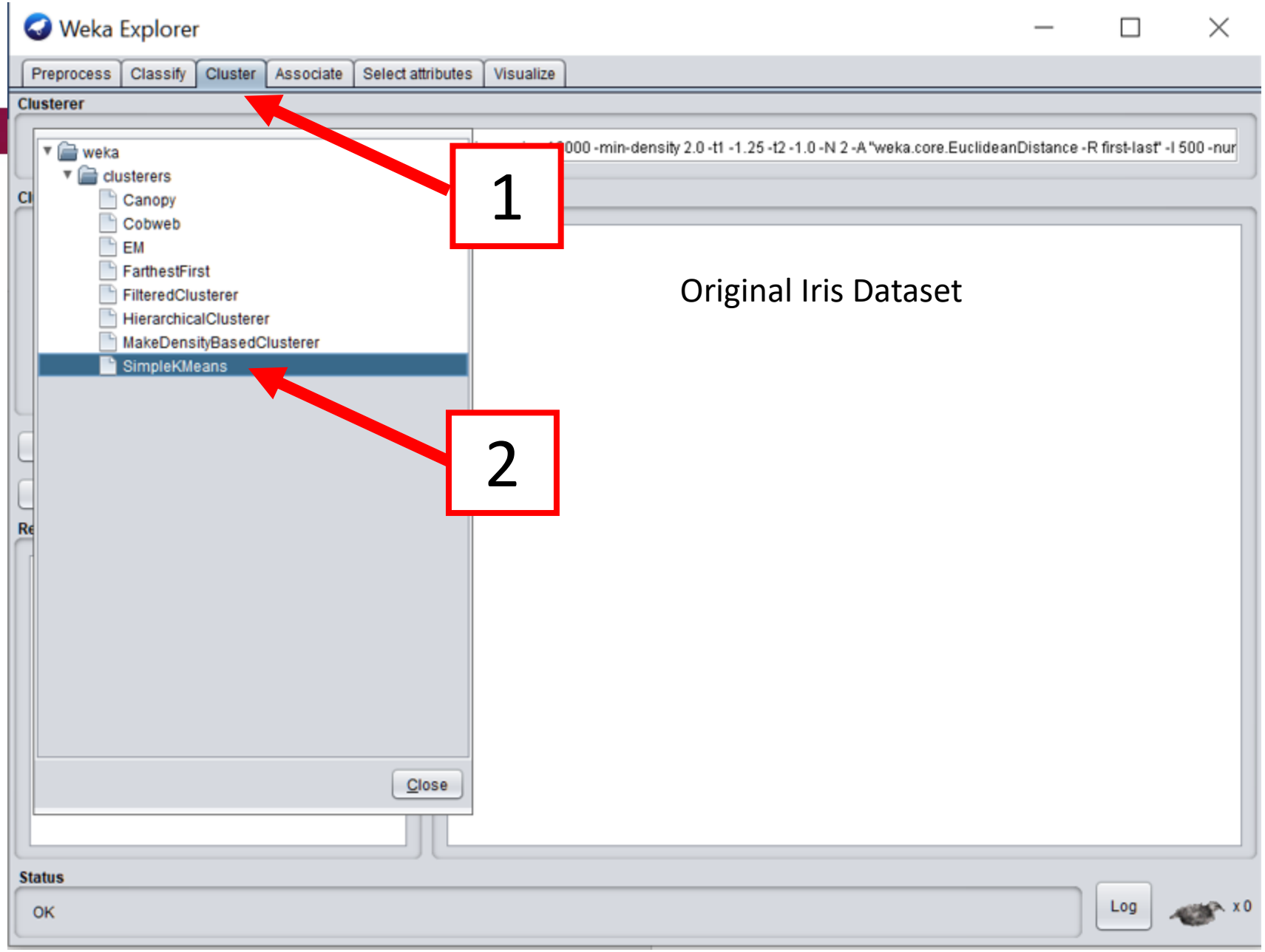


# Introduction to Clustering

- For this lab tutorial, use the original Iris dataset, due to the performance measure for the all three are the same.
- It is based on the dataset either use normalization, standardization and discretization




The best result is using original iris Dataset, iris Dataset with normalization, iris dataset with standardization which has 97.8 (accuracy)




# Step 1

Choose Cluster tab

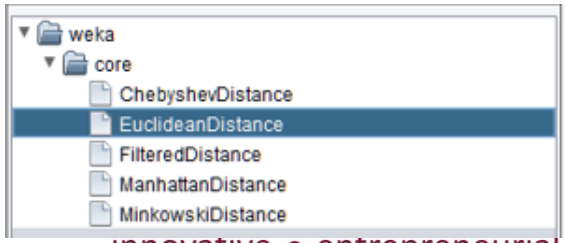
Click Choose>Clusters> SimpleK-Means

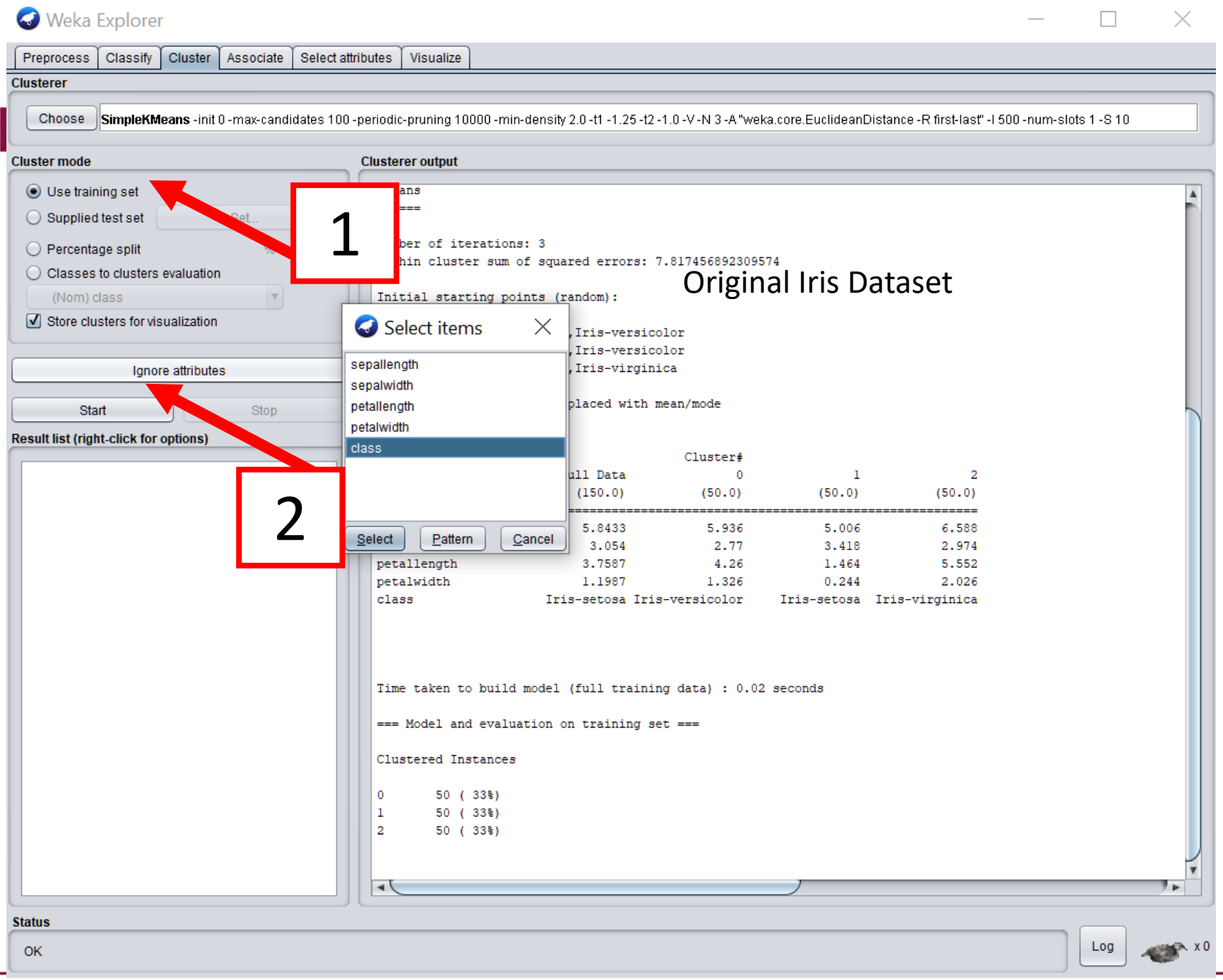
# Step 2

numClusters – the number of desired clusters; set it to 3 because the dataset have three class

displayStdDevs – if true, the standard deviation will be Displayed

distanceFunction- default setting EuclidianDistance





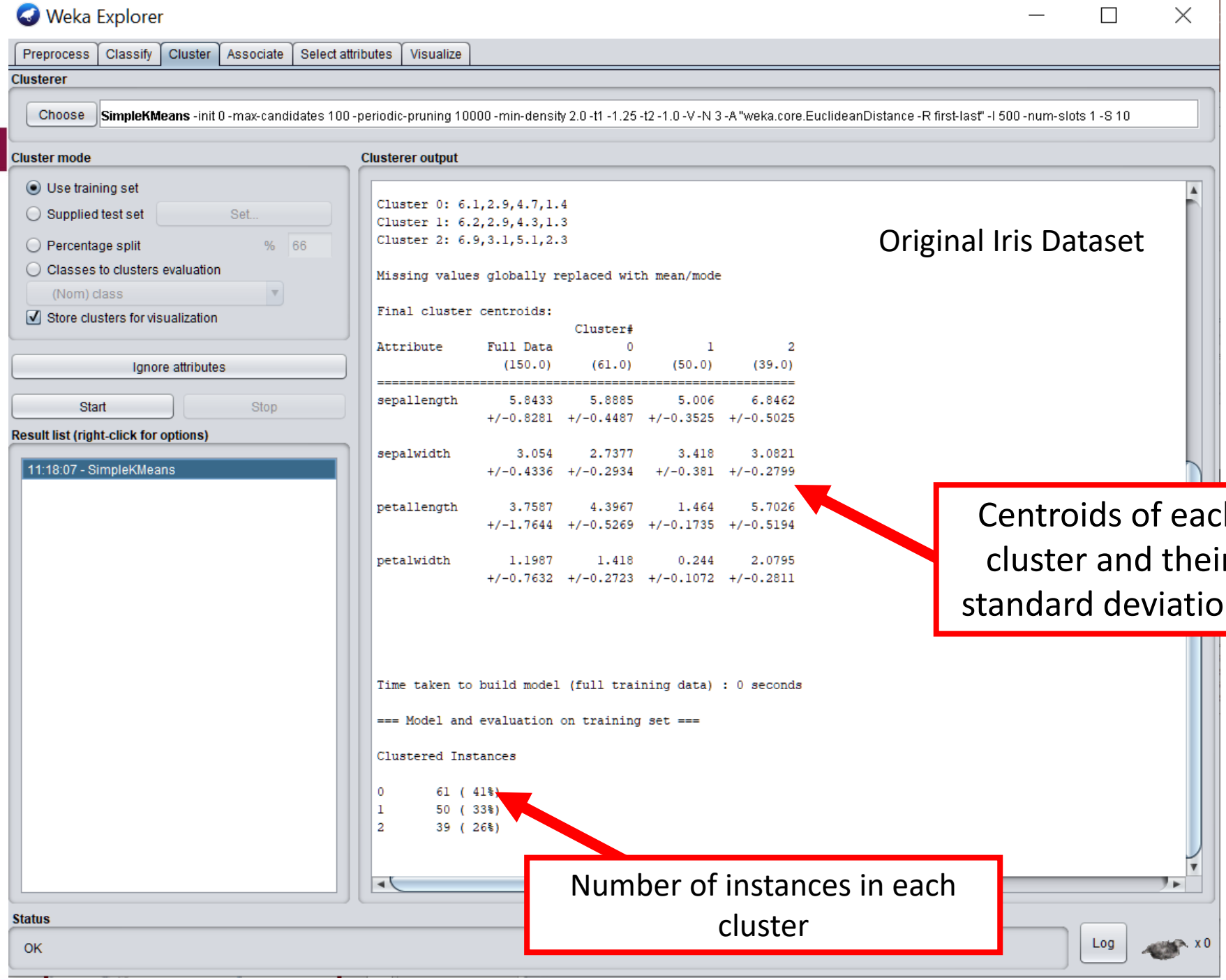

# Step 3

Cluster mode-choose Use training set

Click Ignore attributes. Ignore the Class Attribute.

Click Select

Click Start



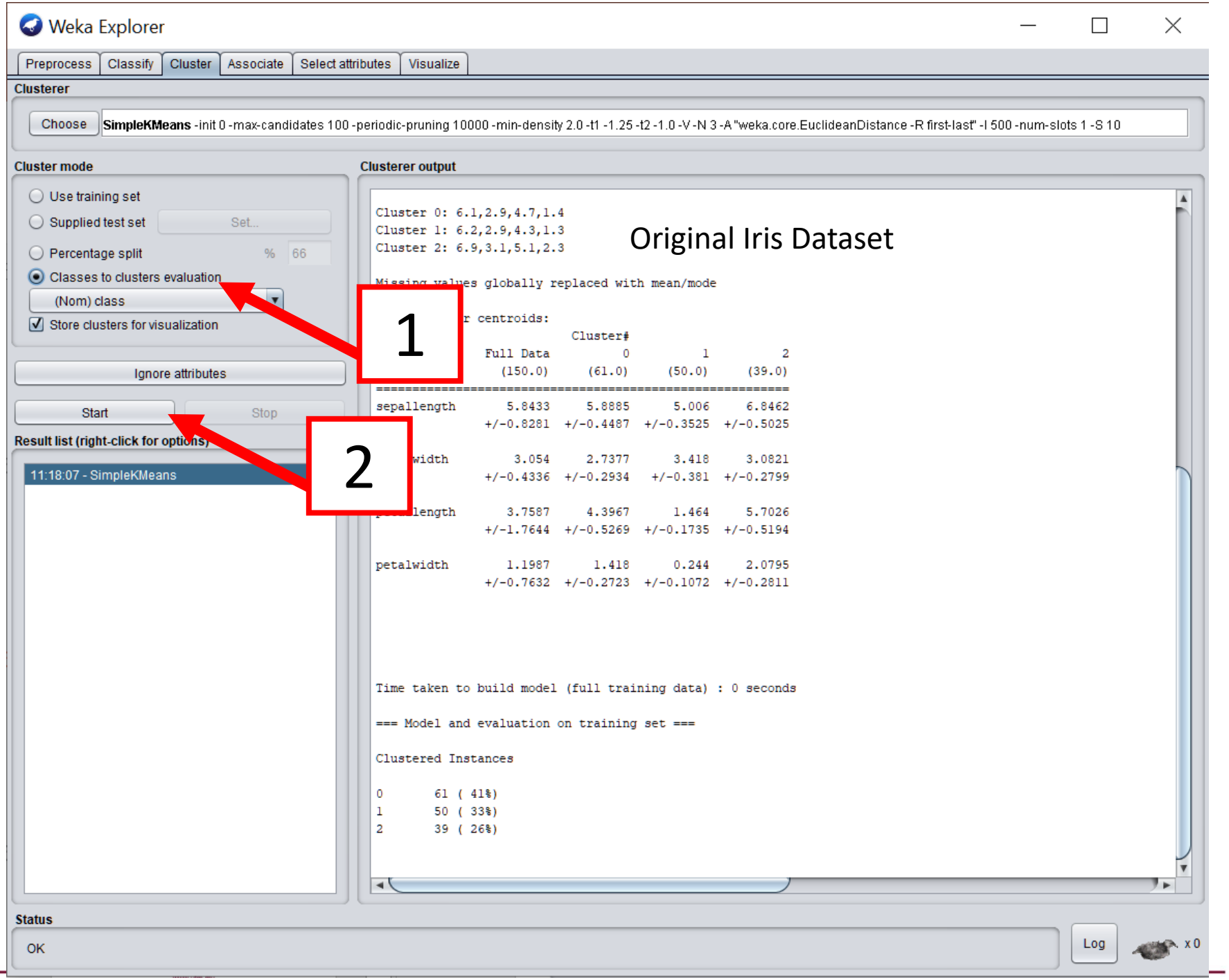

Original Iris Dataset

Centroids of each cluster and their standard deviations

Number of instances in each cluster

# Step 4

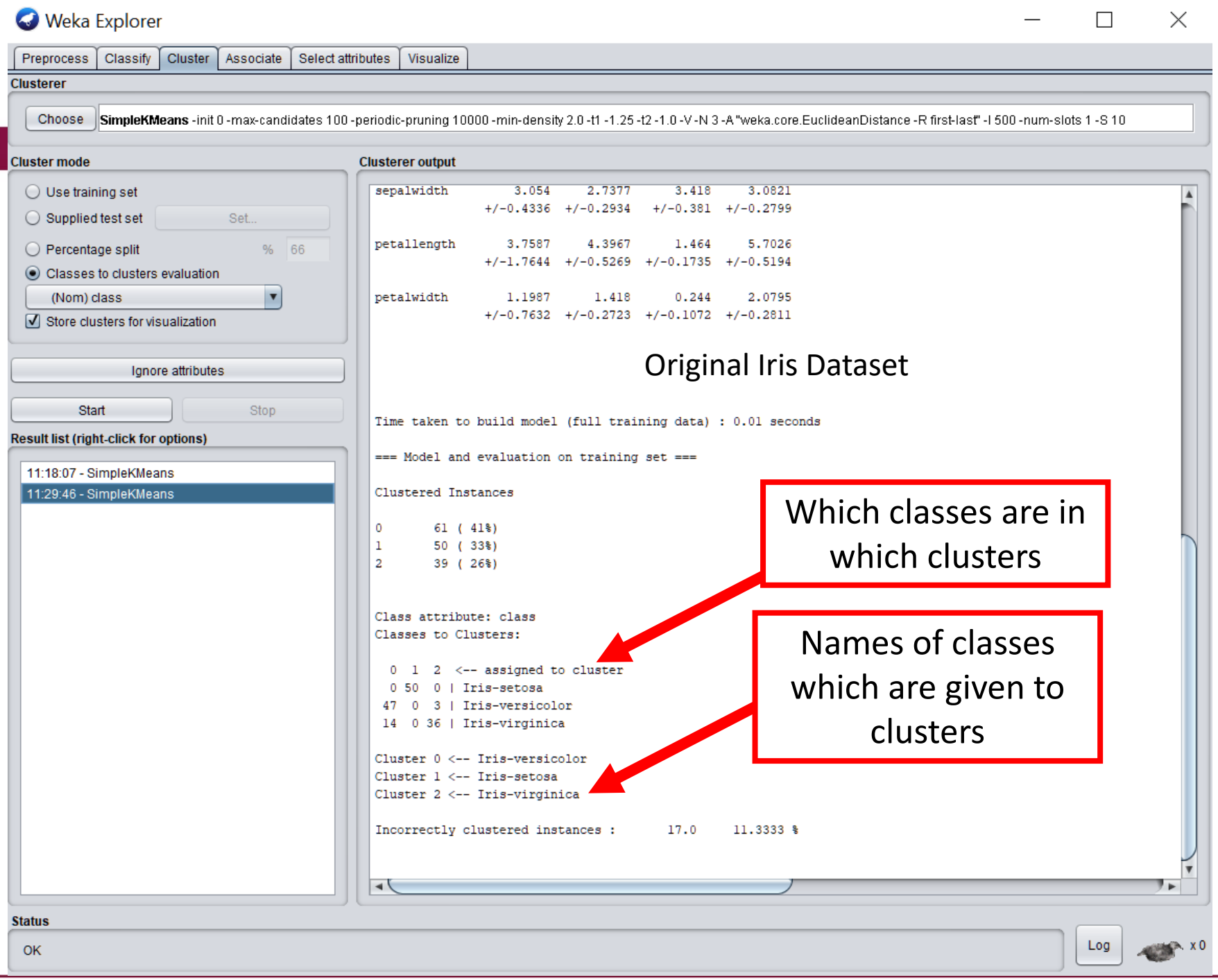
Result using K-Means Clustering




# Step 5

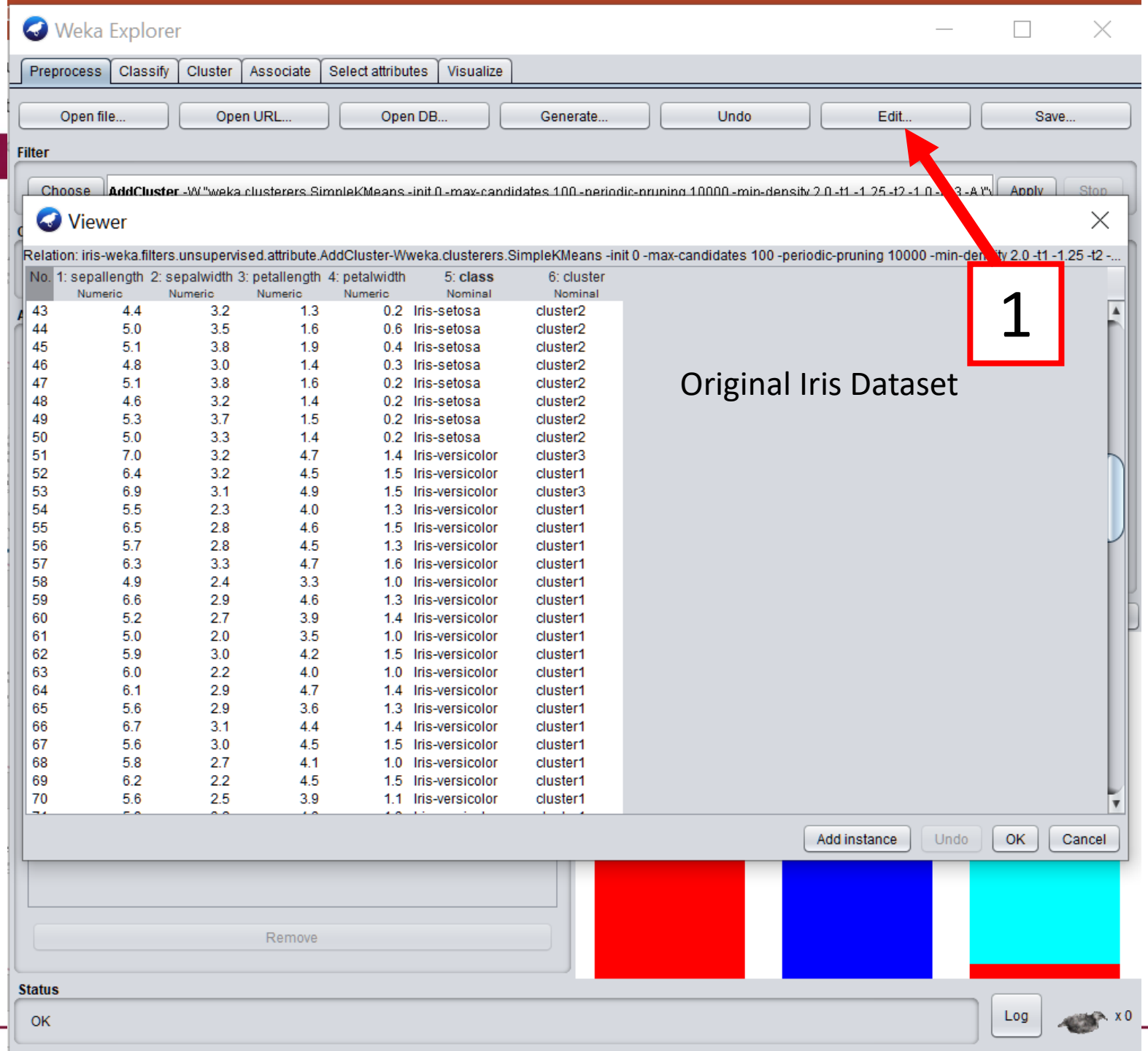
Evaluation of Results

Select the attribute which want to compare the results with.



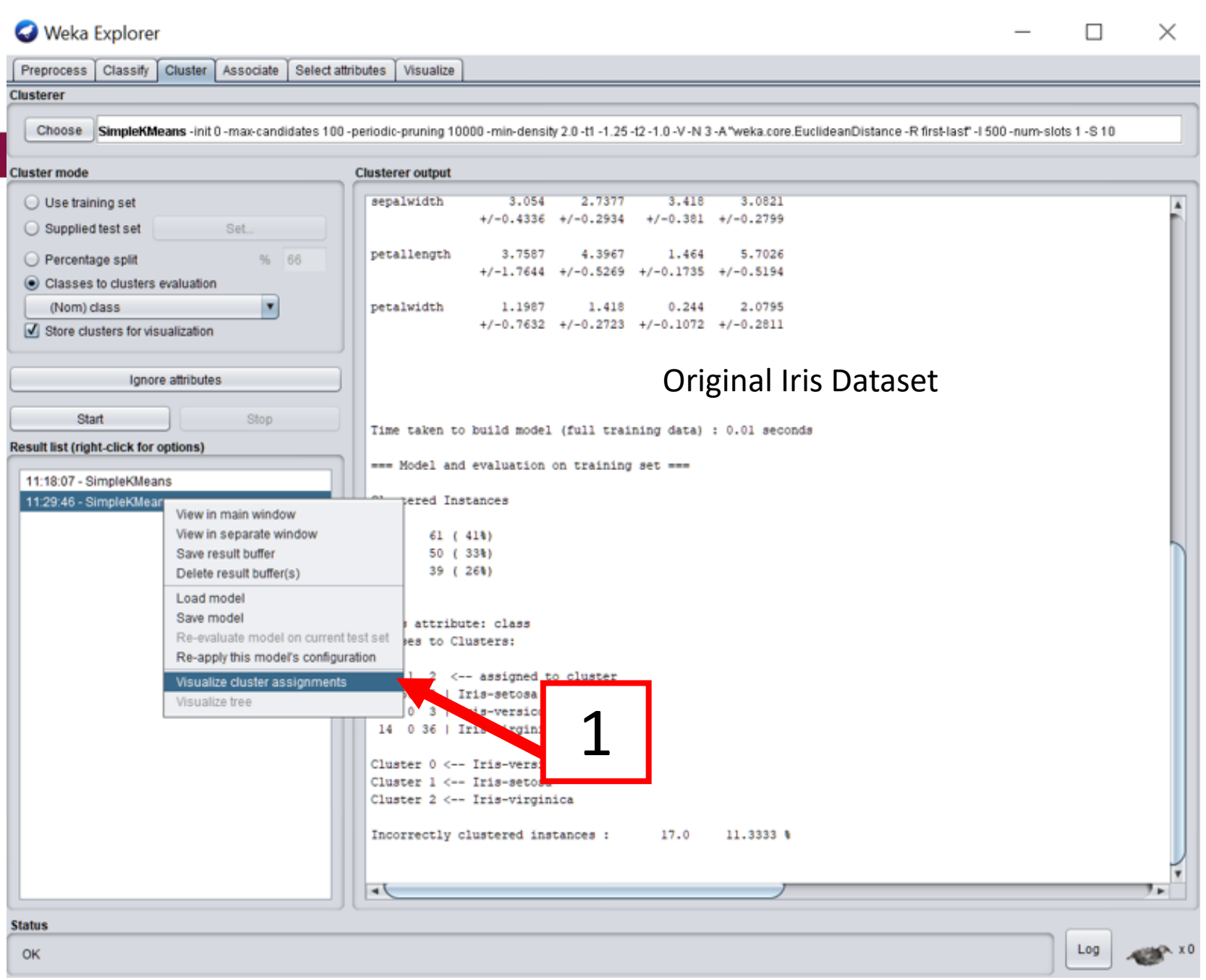

# Step 6

Evaluation of Results




# Step 7

- To further analyze this clustering process, compare the values of cluster attribute (added by the clustering algorithm) with the already existing Class column, by clicking the Edit button.
- 17 samples that Weka clustered differently than the species given in the original dataset.
- Results for the cluster assignments can be added to the dataset.




# Step 8

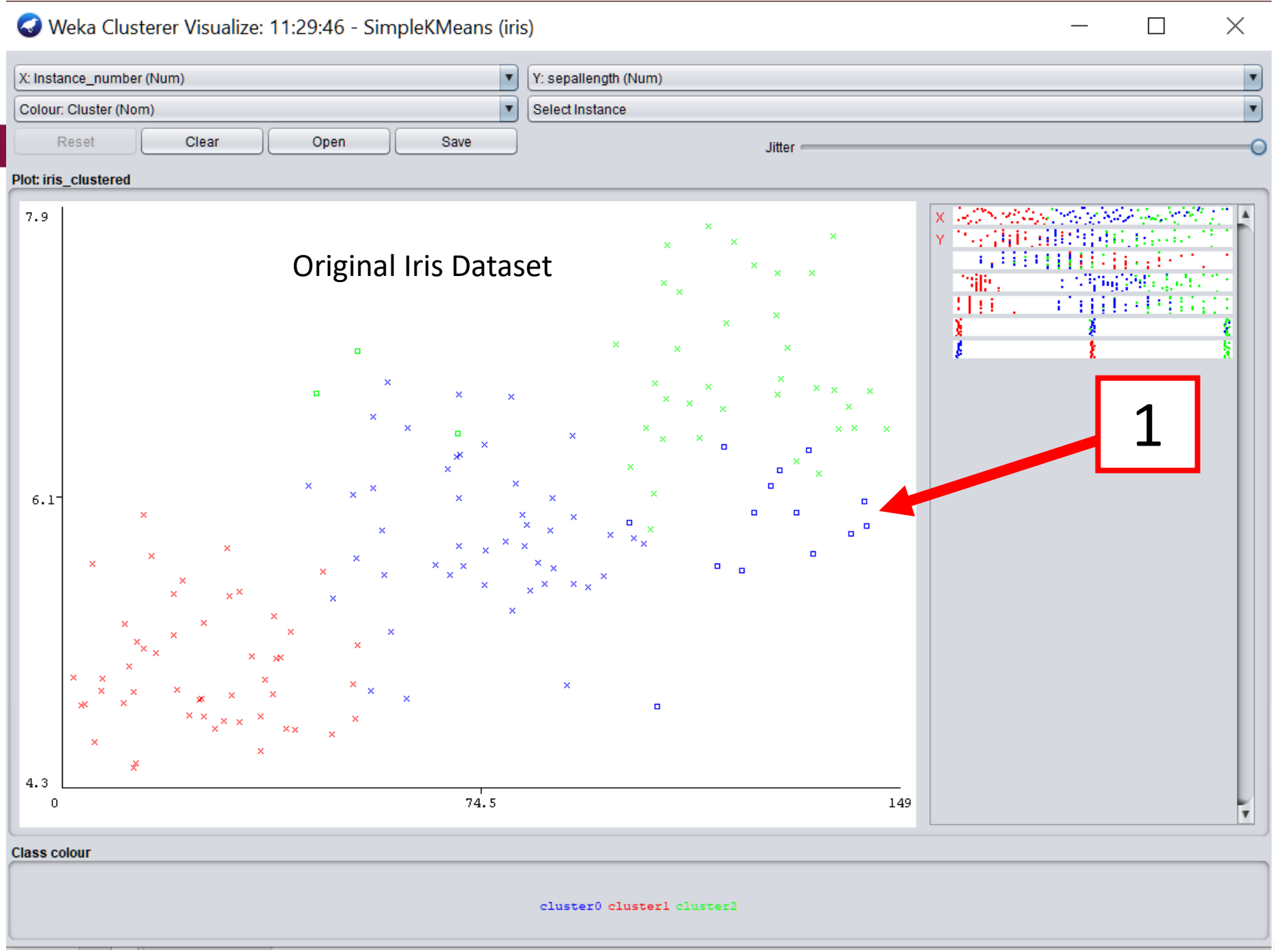
Visual representation of clusters

Right –click for the “Visualize cluster assignments”



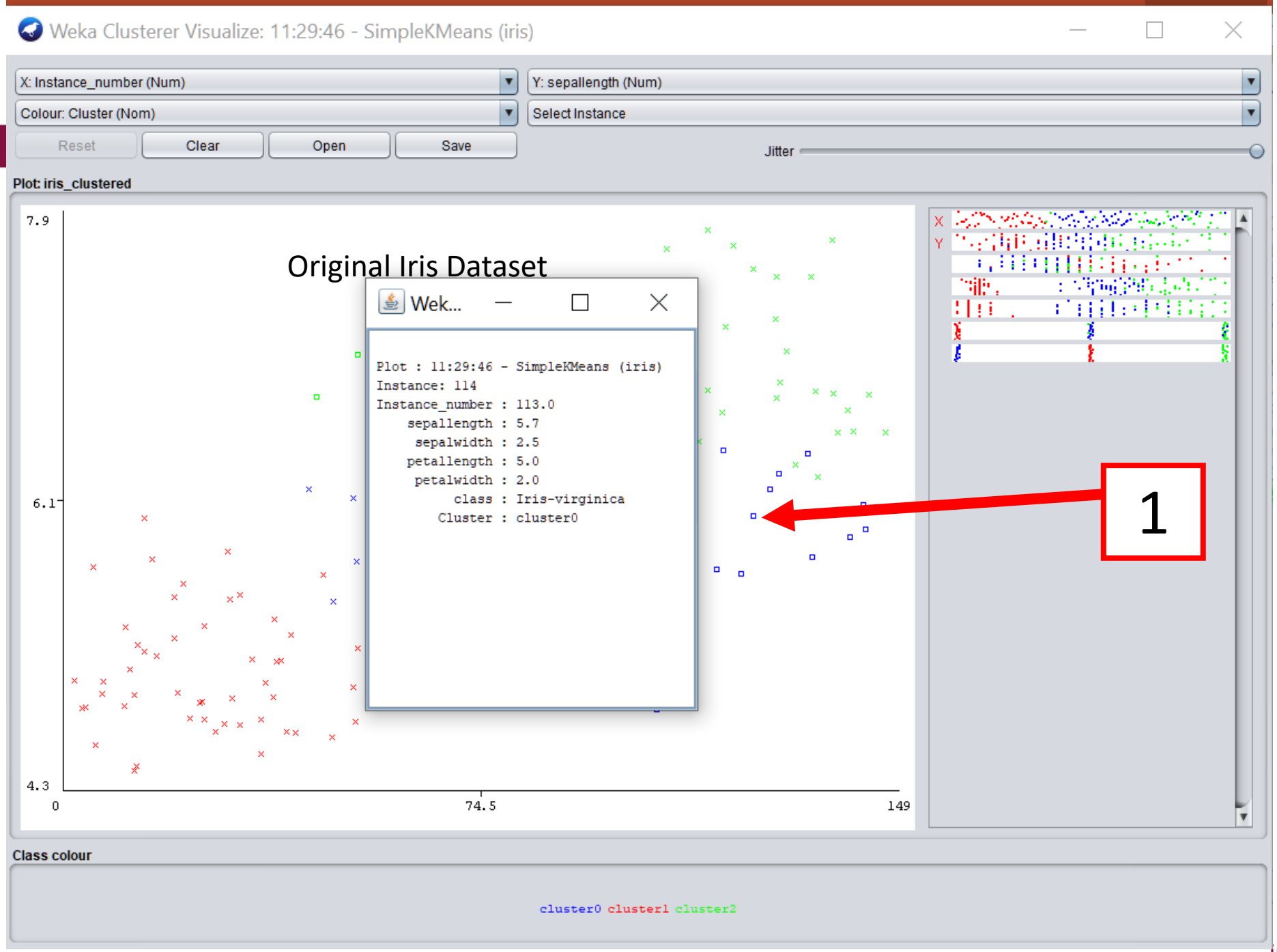

# Step 9

Set the jitter for better visualization




# Step 10

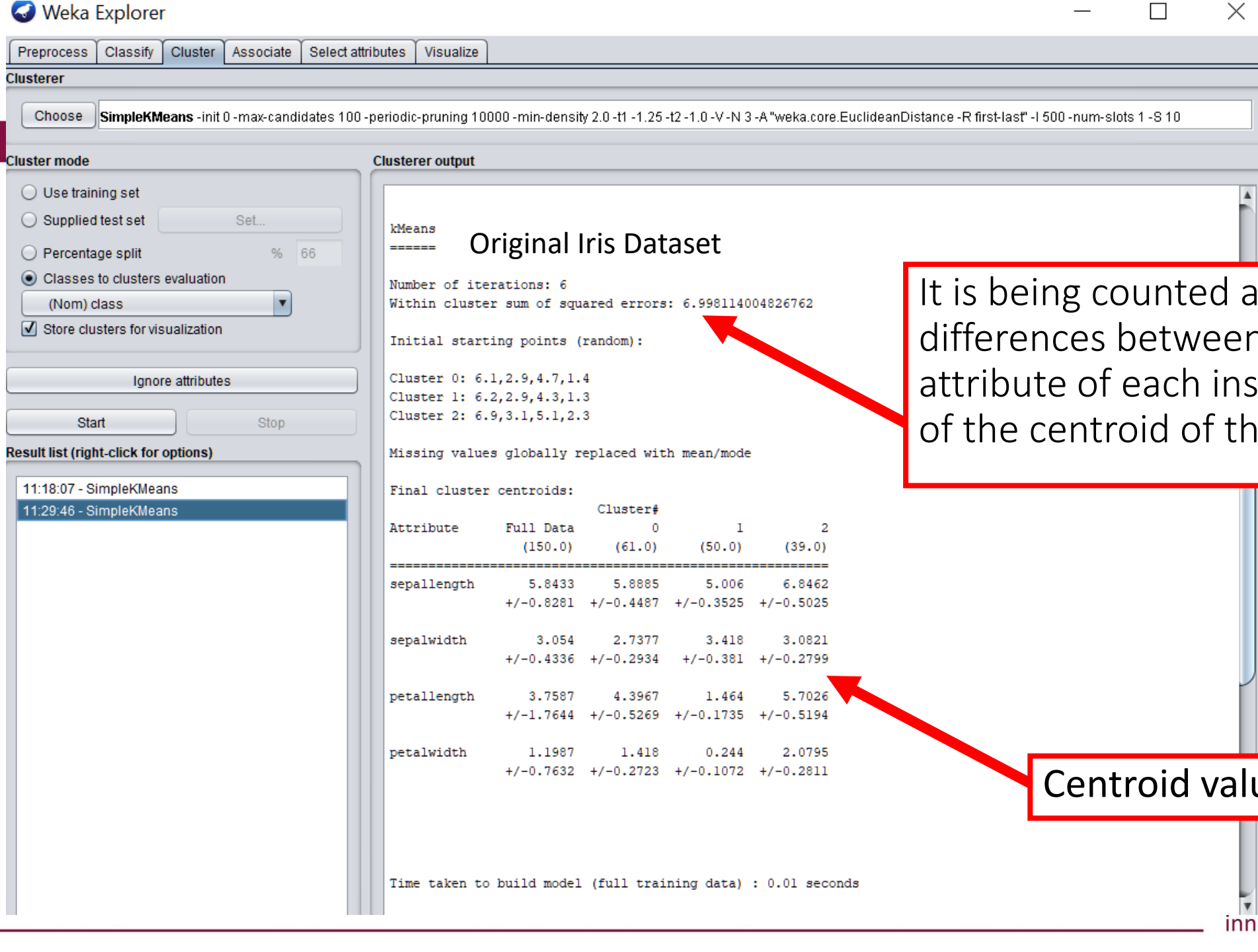
The ▫ shape show the error on the predicted class




# Step 11

Click ▫ shape and detail of the instance will appear

Instance 113 has been misclassified as Iris-setosa




It is being counted as the sum of square differences between the value of the attribute of each instance and the value of the centroid of the given attribute

Centroid values by to all attributes